

Array DSA Question Patterns

- **Elementary Traversal and Modification**

- Sum of elements
- Finding minimum/maximum
- Counting occurrences of an element
- Reversing an array
- Rotate an array

- **Sorting and Searching**

- Sorting algorithms (Bubble, Selection, Insertion, Merge, Quick, Heap)
- Binary search on sorted arrays
- Counting sort (for special cases where elements are in a limited range)
- Search for a given element (linear and binary search)
- Find the kth smallest/largest element

- **Two-pointer Techniques**

- Finding pairs with a specific sum (e.g., two sum problem)
- Reversing an array in-place
- Merging two sorted arrays
- Partitioning an array (used in quicksort)
- Sliding window problems (e.g., maximum sum subarray of size k)

- **Dynamic Programming**

- Maximum/Minimum sum subarray
- Longest increasing subsequence
- Coin change problem
- Knapsack problem variants
- Number of ways to sum up to a target (subset sum problem)

- **Hashing and Counting**

- Frequency count of elements
- Finding duplicates or missing numbers
- Grouping or categorizing data
- Longest consecutive sequence

- **Divide and Conquer**

- Quick sort
- Merge sort
- Binary search
- Find peak element
- Closest pair of points (using divide and conquer in multidimensional arrays)

- **Matrix Problems**

- Matrix rotation and flipping
- Spiral order traversal
- Searching in a row-wise and column-wise sorted matrix
- Matrix multiplication
- Path finding problems (e.g., number of paths from top-left to bottom-right)

- **Cycle Detection**

- Detecting cycle in an array (Floyd's Tortoise and Hare)
- Find the repeating and missing numbers

- **Miscellaneous Complex Patterns**

- Rearrange array according to given indexes
- Rearrange array in-place with $O(1)$ extra space
- Shuffle an array
- Stock buy and sell to maximize profit (single or multiple transactions)
- Reorder array according to given indexes

- **Stacks and Queues using Arrays**

- Next greater/smaller element
- Implementing a min-stack
- Implementing queues using stacks and vice versa
- Histogram related problems (e.g., largest rectangle in histogram)

- **Graph-related**

- Adjacency matrix representation and operations
- Transitive closure of a graph (using Floyd-Warshall algorithm)
- Shortest path in weighted/unweighted graph